

Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	+5V TEST	Header, 2-Pin	P10	1							
	+5V TRANS TEST	Header, 2-Pin	P8	1							
	+12V IN	Header, 2-Pin	P4	1							
	+12V TEST	Header, 2-Pin	P5	1							
	+VOUT TEST	Header, 2-Pin	P7	1							
14	8MHz		X3	1							
21	16MHz		X1	1							
45	DNP/4.7k	RES	R18, R19	2							
46	DNP/100nF	CAP	C31, C35, C36	3							
47	DNP/150k	RES	R28	1							
56	NT	Net Tie SMD	NT1	1							
62	SW2	Switch button 2 pin	SW1	1							
5	2N7002	Mosfet N	Q1	1	onsemi	2N7002LT1G	Volume Production	Newark	2N7002LT1G	0.255	0.255
27	24LCD4BT-E/SN		U14	1	Microchip	24LCD4BT-E/SN	Volume Production	Arrow Electronics	24LCD4BT-E/SN	0.4058	0.4058
67	UART_CONSOLE		P3	1	Molex	53261-0471	Volume Production	Mouser	538-53261-0471	0.69	0.69
68	UART_DEBUG		P2	1	Molex	53261-0471	Volume Production	Mouser	538-53261-0471	0.69	0.69
43	532610671		P6, P9	2	Molex	53261-0671	Volume Production	Newark	53261-0671	0.766	1.53
51	Led	Typical LED	D3, D4, D5, D6, D7, D8, D9	7	Würth Elektronik Midcom	1500608575000	Volume Production	Mouser	710-1500608575000	0.17	1.19
59	Shield		H3, H4, H5, H6	4	Würth Elektronik	9774060360R	Volume Production	Mouser	710-9774060360R	1.75	7
50	Header 4	Header, 4-Pin	JTAG1, P1	2	Würth Elektronik	61300411121	Volume Production	Mouser	710-61300411121	0.19	0.38
44	ATMEGA128A-AU		U6	1	Atmel	ATMEGA128AAU	Volume Production	Newark	ATMEGA128A-AU	7.08	7.08
49	FB 600R	BLM18KG601SN1D	L4	1	Murata	BLM18KG601SN1D	Volume Production	Mouser	81-BLM18KG601SN1D	0.1	0.1
2	1uF	CAP	C3, C4, C40, C46, C53, C65, C68, C72, C75, C79, C83, C84, C85, C86, C87, C89, C90, C91	18	Samsung Electro-Mechanics	CL10A109KA8NNNC	Volume Production				
11	4.7uF	CAP	C38	1	Samsung Electro-Mechanics	CL10A475K08NNNC	Not Recommended for New Design	Mouser	187-CL10A475K08NNNC	0.15	0.15
34	100nF	CAP	C6, C8, C10, C12, C13, C14, C21, C26, C29, C30, C32, C37, C39, C41, C43, C47, C48, C54, C59, C62, C63, C64, C66, C67, C69, C73, C74, C76, C77, C78, C80, C81, C82, C88, C92, C93, C94, C95, C98, C100, C103, C106, C109, C111, C114	45	Samsung Electro-Mechanics	CL10B104K08NNNC	Volume Production	Avnet	CL10B104K08NNNC	0.00367	14.68
6	2.2nF	CAP	C4	1	Samsung Electro-Mechanics	CL10B222KB8NNNC	Volume Production	Mouser	187-CL10B222KB8NNNC	0.1	0.1
7	3.3nF	CAP	C2	1	Samsung Electro-Mechanics	CL10B332KB8NNNC	Volume Production	Mouser	187-CL10B332KB8NNNC	0.2	0.2
10	4.7nF	CAP	C1, C3, C50	3	Samsung Electro-Mechanics	CL10B472KB8NNNC	Volume Production	Mouser	187-CL10B472KB8NNNC	0.16	0.48
18	10uF	CAP	C51, C102	2	Samsung Electro-Mechanics	CL21A106KAYNNE	Volume Production				
23	20mR	RES	R20	1	Bourns	CRM2512-FX-20R0ELF	Volume Production	Mouser	652-CRM2512FX20R0ELF	0.49	0.49
25	22R	RES	R38, R39, R40, R41, R42, R60, R71, R74, R75, R76	10	Panasonic	ERJ2RKF22R0X	Volume Production	Newark	ERJ2RKF22R0X	0.018	0.18
33	100k	RES	R12, R13	2	Panasonic	ERJ3EKF1003V	Volume Production	Mouser	667-ERJ-3EKF1003V	0.1	0.2
37	102k	RES	R22	1	Panasonic	ERJ3EKF1023V	Volume Production	Mouser	667-ERJ-3EKF1023V	0.1	0.1
38	105k	RES	R1, R2	2	Panasonic	ERJ3EKF1053V	Volume Production	Mouser	667-ERJ-3EKF1053V	0.1	0.2
39	120R	RES	R37, R48	2	Panasonic	ERJ3EKF1200V	Volume Production	Newark	ERJ3EKF1200V	0.022	0.22
3	1.33k	RES	R8	1	Panasonic	ERJ3EKF1331V	Volume Production	Mouser	667-ERJ-3EKF1331V	0.1	0.1
28	24.9k	RES	R16, R17	2	Panasonic	ERJ3EKF2492V	Volume Production	Newark	ERJ3EKF2492V	0.099	0.198
8	3.32k	RES	R5, R6	2	Panasonic	ERJ3EKF3321V	Volume Production	Mouser	667-ERJ-3EKF3321V	0.1	0.2
41	332k	RES	R23	1	Panasonic	ERJ3EKF3323V	Volume Production	Mouser	667-ERJ-3EKF3323V	0.1	0.1
9	4.7k	RES	R30, R33, R34, R35, R36, R43, R44, R62, R63	9	Panasonic	ERJ3EKF4701V	Volume Production	Mouser	667-ERJ-3EKF4701V	0.024	0.24
13	5.1k	RES	R66, R69	2	Panasonic	ERJ3EKF5101V	Volume Production	Newark	ERJ3EKF5101V	0.196	1.96
1	1k	RES	R56	1	Panasonic	ERJ-3EKF1001V	Volume Production	Newark	ERJ3EKF1001V	0.099	0.099
16	10k	RES	R3, R4, R9, R10, R11, R26, R31, R32, R45, R46, R47, R49, R52, R54, R57, R58, R59, R61, R64, R65	20	Panasonic	ERJ-3EKF1002V	Volume Production	Mouser	667-ERJ-3EKF1002V	0.024	0.48
19	11.3k	RES	R29	1	Panasonic	ERJ-3EKF1132V	Volume Production	Mouser	667-ERJ-3EKF1132V	0.1	0.1
4	1.5k	RES	R67, R68, R70, R71, R72	5	Panasonic	ERJ-3EKF1501V	Volume Production	Mouser	667-ERJ-3EKF1501V	0.024	0.24
22	20k	RES	R14, R15	2	Panasonic	ERJ-3EKF2002V	Volume Production	Mouser	667-ERJ-3EKF2002V	0.00437	21.85
29	43k	RES	R25	1	Panasonic	ERJ-3EKF4302V	Volume Production	Mouser	667-ERJ-3EKF4302V	0.1	0.1
31	49.9k	RES	R27	1	Panasonic	ERJ-3EKF4992V	Volume Production	Mouser	667-ERJ-3EKF4992V	0.1	0.1
42	665R	RES	R7	1	Panasonic	ERJ-3EKF6650V	Volume Production	Newark	ERJ3EKF6650V	0.099	0.099
48	ESQ-126-14-G-D		H1, H2	2	Samtec	ESQ-126-14-G-D-LL	Volume Production				
20	12pF	CAP	C58, C60, C70, C71	4	Murata	GJM1555C1H120FB01D	Volume Production	Mouser	81-GJM1555C1H120FB01D	0.13	0.52
35	100pF	CAP	C49	1	Murata	GRM1885C1H101JA01D	Not Recommended for New Design	Mouser	81-GRM39C101J50	0.11	0.11
24	22pF	CAP	C56, C57	2	Murata	GRM1885C1H220JA01D	Not Recommended for New Design	Mouser	81-GRM39C220J50	0.11	0.22
	8.2uH	Inductor	L1, L2	2	Vishay	JHLP2525BDER8R2M01	Volume Production	Arrow Electronics	JHLP2525BDER8R2M01	0.7176	1.44
26	22uF	CAP	C5, C7, C9, C11, C15, C16, C19, C20, C24, C25, C27, C28, C42, C44, C51, C52, C97, C99, C105, C108, C110, C113	22	KYOCERA AVX	KG2M21AR51E226MU	Volume Production	Mouser	581-KGM21AR51E226MU	0.653	14.37
69	Shunt 4 Pin	Shunt	R50, R51, R53, R55	4	Susumu	KRL1632E-M-R010-F-T5	Volume Production	DigiKey	408-1570-1-ND	0.43	1.72
52	LMR33630ADDA		U3, U4	2	Texas Instruments	LMR33630ADDA	Volume Production	Mouser	595-LMR33630ADDA	2.89	5.78
54	MAX485ESA+		U9	1	Analog Devices	MAX485ESA+	Volume Production	Newark	MAX485ESA+	6.41	6.41
55	MCP2515-SO	Stand-Alone CAN Controller with SPI Interface	U7	1	Microchip	MCP2515-1-SO	Volume Production	Arrow Electronics	MCP2515-1/SO	2.39	2.39
70	Crystal 2 Pins	Crystal	X2	1	NDK	NX2012SA-32.768K-STD-MUB-1		Digi-Key	644-1219-1-ND		
57	PAC1934T-I/JQ		U10	1	Microchip	PAC1934T-I/JQ	Volume Production	Arrow Electronics	PAC1934T-I/JQ	1.88	1.88
58	PCA9508DP,118		U15	1	NXP Semiconductors	PCA9508DP,118	Volume Production	Newark	PCA9508DP,118	2.32	2.32
17	10mR	RES	R21, R24	2	Yageo Group	RL0603FR-070R01L	Volume Production	Mouser	660-RL0603FR-070R01L	0.46	0.92
60	SMBJ16A	TVS Diode	D1, D2, D10	3	Taiwan Semiconductor	SMBJ16A	Volume Production	Mouser	821-SMBJ16A	0.61	1.83
	SN65C3232PWR		U12	1	Texas Instruments	SN65C3232PWR	Volume Production	Mouser	595-SN65C3232PWR	3.31	3.31
61	SN65HVD251DR		U8	1	Texas Instruments	SN65HVD251DR	Volume Production	Newark	SN65HVD251DR	4.15	4.15
12	4.7uH	Inductor	L3	1	Bourns	SRP65404R7M	Volume Production	Mouser	652-SRP6540-4R7M	1.16	1.16
40	120uF	T543X107M025AHE060	C45	1	Yageo Group	T543X107M025AHE060	Volume Production	Mouser	80-T543X107M025AHE060	14.67	14.67
32	68uF	T543X336M050AHE04075 05	C55	5	Yageo Group	T543X336M050AHE040	Volume Production	Mouser	80-T543X336M050AHE040	13.05	13.05
36	100uF	Capacitor pol	C96, C101, C104, C107, C112	5	KYOCERA AVX	TAJE107K016RNJ	Volume Production	Arrow Electronics	TAJE107K016RNJ	1.48	7.42
63	TC1047AVNBTR		U11	1	Microchip	TC1047AVNBTR	Volume Production	Arrow Electronics	TC1047AVNBTR	0.7759	0.7759
30	47uF	CAP	C17, C18, C22, C23	4	TAIYO YUDEN	TKM325AB3476MMP	Volume Production	Mouser	963-TMK325AB3476MMP	1.21	4.84
64	TLR5010QDDCQ1		U13	1	Texas Instruments	TLR5010QDDCQ1	Volume Production	Mouser	595-TLR5010QDDCQ1	1.79	1.79
65	TPS53287RYQR		U5	1	Texas Instruments	TPS53287RYQR	Volume Production	Mouser	595-TPS53287RYQR	5.32	5.32
66	TPS25947OARPWR		U1, U2	2	Texas Instruments	TPS25947OARPWR	Volume Production	Mouser	595-TPS25947OARPWR	1.72	3.44